1.14 Polynomials

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Terminology

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$$11 \div 4 = 2rem3$$

dividend
(the thing we're dividing)

divisor
(the thing we're dividing by)

quotient

remainder

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Normal Long Division

7

$$\begin{array}{r|l} 11 & 423.0000 \end{array}$$

1. We found how many whole number of times (i.e. the quotient) the divisor went into the dividend.

2. We multiplied the quotient by the dividend.

3. …in order to find the remainder.

4. Find we 'brought down' the next number.

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Example

8

How many times does x go into $6x^3$?

$$6x^2 - 2x + 3$$

Now repeat! How many times does x go into $-2x^2$?

$$x + 5 \;\overline{)\; 6x^3 + 28x^2 - 7x + 15}$$

Multiply $6x^2$ by $(x + 5)$. The first term should match with above.

$$6x^3 + 30x^2$$

Subtract and carry down next term.

$$-2x^2 - 7x$$

$$-2x^2 - 10x$$

$$3x + 15$$

$$3x + 15$$

This is the remainder.

$$0$$

Tip:
You can check your solution by expanding:
$(x + 5)(6x^2 - 2x + 3)$
But if you know you should get no remainder, ending with 0 at the bottom is a good sign!

Tip:
Be **very** careful subtracting negatives.
$-7x - (-10x) = 3x$

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Further Example

9

Find the remainder when $3x^3 - 2x + 4$ is divided by $x - 1$.

$$x - 1 \;\overline{)\; 3x^3 + 0x^2 - 2x + 4}$$

Note: Ensure that any missing terms in the polynomial are filled in, so that the powers decrease by 1 each time.

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Test Your Understanding

10

Find the remainder when $2x^3 - 5x^2 - 16x + 10$ is divided by $x - 4$.

$$x - 4 \quad 2x^3 - 5x^2 - 16x + 10$$

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Further Test Your Understanding

11

Divide $8x^3 - 1$ by $2x - 1$.

$$2x - 1 \quad 8x^3 + 0x^2 + 0x - 1$$

Fro Fun Fact: There is a well-known factorisation for the '*difference of two cubes*':
$x^3 - y^3 = (x - y)\left(x^2 + xy + y^2\right)$
This is NOT in the A Level syllabus.

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Exercise 7B – Y1 Pages 142

12

15 $\mathrm{f}(x) = 10x^3 + 43x^2 - 2x - 10$.
Find the remainder when $\mathrm{f}(x)$ is divided by $(5x + 4)$.

16 $\mathrm{f}(x) = 3x^3 - 14x^2 - 47x - 14$.
a Find the remainder when $\mathrm{f}(x)$ is divided by $(x - 3)$.
b Given that $(x + 2)$ is a factor of $\mathrm{f}(x)$, factorise $\mathrm{f}(x)$ completely.

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Exercise 7B – Y1 Pages 142

13

17 a Find the remainder when $x^3 + 6x^2 + 5x - 12$ is divided by
i $x - 2$,
ii $x + 3$.
b Hence, or otherwise, find all the solutions to the equation $x^3 + 6x^2 + 5x - 12 = 0$.

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Exercise 7B – Y1 Pages 142

14

18 $\mathrm{f}(x) = 2x^3 + 3x^2 - 8x + 3$.
a Show that $\mathrm{f}(x) = (2x - 1)(ax^2 + bx + c)$ where a, b and c are constants to be found.
b Hence factorise $\mathrm{f}(x)$ completely.
c Write down all the real roots of the equation $\mathrm{f}(x) = 0$.

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Exercise 7B – Y1 Pages 142

15

19 $\mathrm{f}(x) = 12x^3 + 5x^2 + 2x - 1$.
a Show that $(4x - 1)$ is a factor of $\mathrm{f}(x)$ and write $\mathrm{f}(x)$ in the form $(4x - 1)(ax^2 + bx + c)$.
b Hence, show that the equation $12x^3 + 5x^2 + 2x - 1 = 0$ has exactly 1 real solution.

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3: The Factor Theorem

16

$$x^3 + x^2 - 4x - 4 = (x - 2)\left(x^2 + 3x + 2\right)$$

We can see that $(x - 2)$ is a factor of $x^3 + x^2 - 4x - 4$.
What would happen if x is 2?

The converse is also true: if we could find a value a such that the LHS is 0 when we substitute in a for x, then $(x - a)$ would be a factor.

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Examples (in textbook)

17

Show that $(x - 2)$ is a factor of $x^3 + x^2 - 4x - 4$.

Let $f(x) = x^3 + x^2 - 4x - 4$

Writing $f(x) = \ldots$ gives us appropriate notation, i.e. $f(2)$, to show we're substituting 2 into the polynomial on the next line.

Fully factorise $2x^3 + x^2 - 18x - 9$.

Let $f(x) = 2x^3 + x^2 - 18x - 9$

Keep on trying values until you find one where $f(p) = 0$. Recommended order: $p = 1, -1, 2, -2, \ldots$
You can use the 'Table' mode on your calculator to try lots of values in a range – see my interactive Classwiz powerpoint.



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Using Factor Theorem to find unknown coefficients

18

Given that $2x + 1$ is a factor of $6x^3 + ax^2 + 1$, determine the value of a.

Notes:
Choose the value of x to substitute that **makes the divisor/factor 0**.
So if $2x + 1 = 0$, then $x = -\frac{1}{2}$, thus find $f\left(-\frac{1}{2}\right)$.
This even works when you're dividing by non-linear factors, e.g. $\frac{1}{x} + 1$, but at A Level they will always be of the form $ax + b$.

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Test Your Understanding

19

Edexcel C2 May 2016 Q2

$\mathrm{f}(x) = 6x^3 + 13x^2 - 4$

No long in spec.

(a) Use the remainder theorem to find the remainder when $\mathrm{f}(x)$ is divided by $(2x + 3)$. \hfill(2)
~~(b) Use the factor theorem to show that $(x + 2)$ is a factor of $\mathrm{f}(x)$. \hfill(2)~~
(c) Factorise $\mathrm{f}(x)$ completely. \hfill(4)

Given that $3x - 1$ is a factor of $3x^3 + 11x^2 + ax + 1$, determine the value of a.

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Exercise 7C – Y1 Pages 146

20

13 $\mathrm{f}(x) = 3x^3 - 12x^2 + 6x - 24$.
a Use the factor theorem to show that $(x - 4)$ is a factor of $\mathrm{f}(x)$.
b Hence, show that 4 is the only real root of the equation $\mathrm{f}(x) = 0$.

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Exercise 7C – Y1 Pages 145-146

21

14 $\mathrm{f}(x) = 4x^3 + 4x^2 - 11x - 6$.
a Use the factor theorem to show that $(x + 2)$ is a factor of $\mathrm{f}(x)$.
b Factorise $\mathrm{f}(x)$ completely.

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Exercise 7C – Y1 Pages 145-146

22

15 a Show that $(x-2)$ is a factor of $9x^4 - 18x^3 - x^2 + 2x$.
b Hence, find four real solutions to the equation $9x^4 - 18x^3 - x^2 + 2x = 0$.

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Exercise Extension

23

1 *[MAT 2006 1E]* The cubic
$$x^3 + ax + b$$
Has both $(x-1)$ and $(x-2)$ as factors. Determine the values of a and b.

[MAT 2009 1I] The polynomial
$$n^2x^{2n+3} - 25nx^{n+1} + 150x^7$$
2 Has $x^2 - 1$ as a factor
A) for no values of n;
B) for $n = 10$ only;
C) for $n = 15$ only;
D) for $n = 10$ and $n = 15$ only.

The **remainder theorem** states that if $f(x)$ is divided by $(x-a)$, the remainder is $f(a)$. This similarly works whenever a makes the divisor 0.

3 [MAT 2013 1G] Let $n \geq 2$ be an integer and $p_n(x)$ be the polynomial
$p_n(x) = (x-1) + (x-2) + \ldots + (x-n)$
What is the remainder, in terms of n, when $p_n(x)$ is divided by $p_{n-1}(x)$?